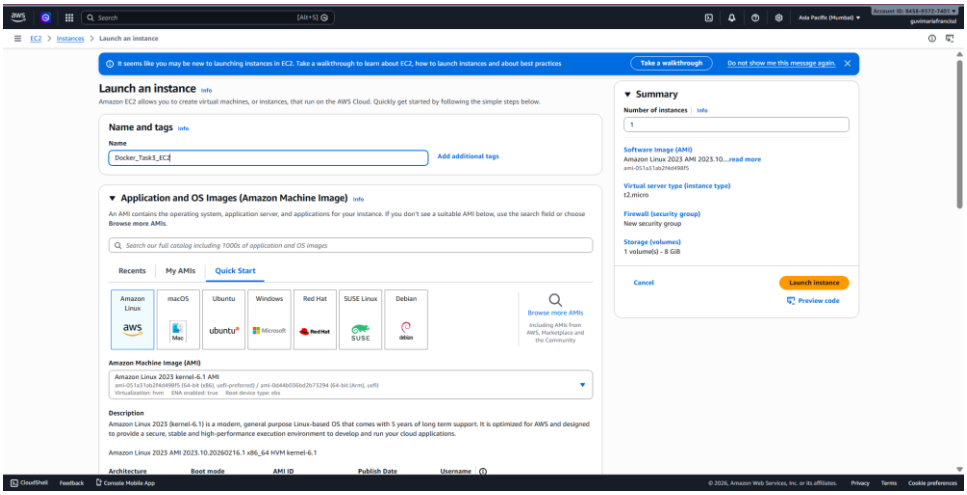
Docker Task -3

Task Description:

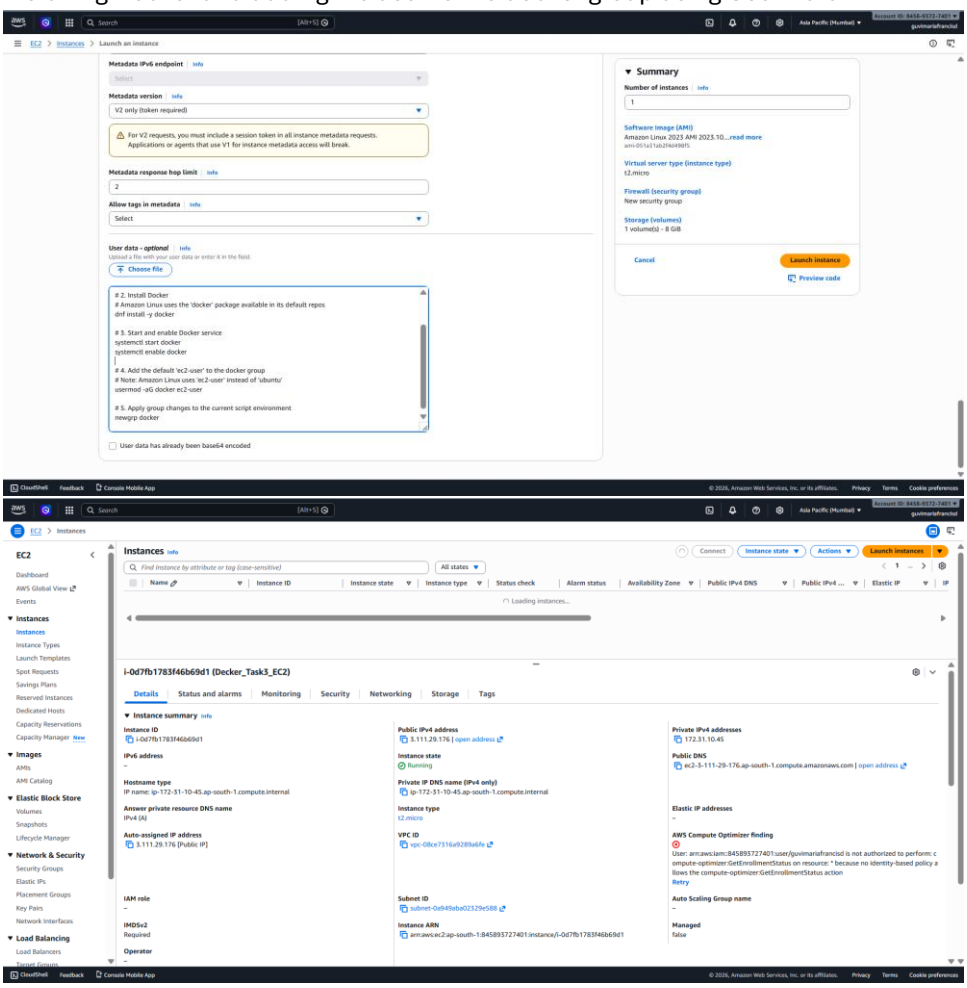
- **Create a custom docker image for nginx and**
- **deploy it using docker compose,**
- **where the volume bind mount should be at /var/opt/nginx location.**
- **Push the created custom docker image to your docker-hub.**

1. Creating a custom docker image for nginx:

Creating an EC2 instance:



Installing Docker and adding the user to the docker group using User Data:



Docker Already Installed and is running:

```
ec2-user@ip-172-31-10-45:~
[ec2-user@ip-172-31-10-45 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-10-45 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-03-03 13:37:13 UTC; 9min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 4775 (dockerd)
      Tasks: 7
     Memory: 28.3M
        CPU: 320ms
     CGroup: /system.slice/docker.service
             └─4775 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Mar 03 13:37:12 ip-172-31-10-45.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Mar 03 13:37:12 ip-172-31-10-45.ap-south-1.compute.internal dockerd[4775]: time="2026-03-03T13:37:12.834136025Z" level=info msg="Starting up"
Mar 03 13:37:12 ip-172-31-10-45.ap-south-1.compute.internal dockerd[4775]: time="2026-03-03T13:37:12.904022961Z" level=info msg="Loading containers: start."
Mar 03 13:37:13 ip-172-31-10-45.ap-south-1.compute.internal dockerd[4775]: time="2026-03-03T13:37:13.440393458Z" level=info msg="Loading containers: done."
Mar 03 13:37:13 ip-172-31-10-45.ap-south-1.compute.internal dockerd[4775]: time="2026-03-03T13:37:13.471181173Z" level=info msg="Docker daemon" commit=d334
Mar 03 13:37:13 ip-172-31-10-45.ap-south-1.compute.internal dockerd[4775]: time="2026-03-03T13:37:13.471299328Z" level=info msg="Daemon has completed initi
Mar 03 13:37:13 ip-172-31-10-45.ap-south-1.compute.internal dockerd[4775]: time="2026-03-03T13:37:13.533702343Z" level=info msg="API listen on /run/docker.
Mar 03 13:37:13 ip-172-31-10-45.ap-south-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
[ec2-user@ip-172-31-10-45 ~]$
```

2. Deploy it using docker compose

Installed Docker Compose and creating the required Files: Dockerfile, webpage and compose file:

```
ec2-user@ip-172-31-10-45:~/r
[ec2-user@ip-172-31-10-45 ~]$ docker compose version
Docker Compose version v5.1.0
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$ ls
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$ mkdir my_nginx_image
[ec2-user@ip-172-31-10-45 ~]$ ls
my_nginx_image
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$ cd my_nginx_image/
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$
```

```
ec2-user@ip-172-31-10-45:~/r
<!DOCTYPE html>
<html>
<head>
<title>Custom Nginx</title>
</head>
<body>
<h1>Custom Nginx Docker Container Running</h1>
<p>Deployed using Docker Compose</p>
</body>
</html>
```

```
ec2-user@ip-172-31-10-45:~/r
version: '1'

services:
  nginx:
    image: ghostatdocker/my_nginx_image:1.0
    container_name: my_nginx_con
    ports:
      - "80:80"
    volumes:
      - ./nginx-data:/var/opt/nginx
```

```
ec2-user@ip-172-31-10-45:~/r
FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
~
```

```
ec2-user@ip-172-31-10-45:~/r
[ec2-user@ip-172-31-10-45 ~]$ ls
[ec2-user@ip-172-31-10-45 ~]$ mkdir my_nginx_image
[ec2-user@ip-172-31-10-45 ~]$ ls
my_nginx_image
[ec2-user@ip-172-31-10-45 ~]$ cd my_nginx_image/
[ec2-user@ip-172-31-10-45 my_nginx_image]$ ls
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$ vi Dockerfile
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$ cat Dockerfile
FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$ vi index.html
[ec2-user@ip-172-31-10-45 my_nginx_image]$ cat index.html
<!DOCTYPE html>
<html>
<head>
<title>Custom Nginx</title>
</head>
<body>
<h1>Custom Nginx Docker Container Running</h1>
<p>Deployed using Docker Compose</p>
</body>
</html>
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$
```

```
ec2-user@ip-172-31-10-45:~/r
[ec2-user@ip-172-31-10-45 my_nginx_image]$ tree ..
..
└── my_nginx_image
    ├── Dockerfile
    └── index.html

1 directory, 2 files
[ec2-user@ip-172-31-10-45 my_nginx_image]$ vi docker-compose.yml
[ec2-user@ip-172-31-10-45 my_nginx_image]$ cat docker-compose.yml
version: '1'

services:
  nginx:
    image: ghostatdocker/my_nginx_image:1.0
    container_name: my_nginx_con
    ports:
      - "80:80"
    volumes:
      - ./nginx-data:/var/opt/nginx

[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$
```

3. **where the volume bind mount should be at /var/opt/nginx location:**

```
[ec2-user@ip-172-31-10-45 my_nginx_image]$ tree ..
.
└── my_nginx_image
    ├── Dockerfile
    ├── docker-compose.yml
    ├── index.html
    └── nginx-data

2 directories, 3 files
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker compose up -d
WARN[0000] /home/ec2-user/my_nginx_image/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
[+] up 2/2
 ✔Network my_nginx_image_default Created                                                                                                    0.1s
 ✔Container my_nginx_con         Started                                                                                                    0.7s
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker images
REPOSITORY                    TAG       IMAGE ID       CREATED          SIZE
ghostatdocker/my_nginx_image   1.0       2428b8a60c9c   25 minutes ago   161MB
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS          PORTS                                      NAMES
e39be9e0fb20   ghostatdocker/my_nginx_image:1.0   "/docker-entrypoint.…"   24 seconds ago   Up 23 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   my_nginx_con
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$ curl ifconfig.me
3.111.29.176[ec2-user@ip-172-31-10-45 my_nginx_image]$
```

Custom Nginx

Not secure 3.111.29.176

Custom Nginx Docker Container Running

Deployed using Docker Compose

Checking the Bind Mount status by creating a file In the mounted volume and verifying the same in EC2 folder:

```
[ec2-user@ip-172-31-10-45 ~]$ ls
my_nginx_image
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS          PORTS                                      NAMES
e39be9e0fb20   ghostatdocker/my_nginx_image:1.0   "/docker-entrypoint.…"   3 minutes ago    Up 3 minutes    0.0.0.0:80->80/tcp, :::80->80/tcp   my_nginx_con
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$ docker exec -it my_nginx_con bash
root@e39be9e0fb20:/# ls
bin  boot  dev  docker-entrypoint.d  docker-entrypoint.sh  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@e39be9e0fb20:/# ls /var/opt/nginx/
root@e39be9e0fb20:/# cd /var/opt/nginx/
root@e39be9e0fb20:/var/opt/nginx#
root@e39be9e0fb20:/var/opt/nginx# touch FileCreatedinContainer
root@e39be9e0fb20:/var/opt/nginx#
root@e39be9e0fb20:/var/opt/nginx# exit
exit
[ec2-user@ip-172-31-10-45 ~]$
[ec2-user@ip-172-31-10-45 ~]$ cd my_nginx_image
[ec2-user@ip-172-31-10-45 my_nginx_image]$ ls
Dockerfile  docker-compose.yml  index.html  nginx-data
[ec2-user@ip-172-31-10-45 my_nginx_image]$ cd nginx-data/
[ec2-user@ip-172-31-10-45 nginx-data]$ ls
FileCreatedinContainer
[ec2-user@ip-172-31-10-45 nginx-data]$
[ec2-user@ip-172-31-10-45 nginx-data]$
```

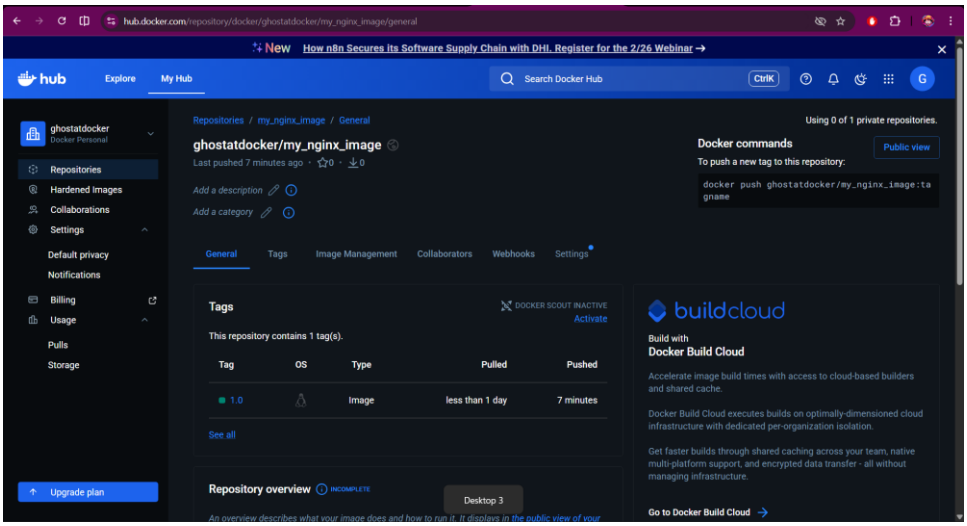
```
ec2-user@ip-172-31-10-45:~

[ec2-user@ip-172-31-10-45 ~]$ tree
.
└── my_nginx_image
    ├── Dockerfile
    ├── docker-compose.yml
    ├── index.html
    └── nginx-data
        └── FileCreatedinContainer

2 directories, 4 files
[ec2-user@ip-172-31-10-45 ~]$
```

4. Building a Custom Nginx Docker Image and pushing it to Docker Hub:

```
ec2-user@ip-172-31-10-45:~/i
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker build -t ghostatdocker/my_nginx_image:1.0 .
[+] Building 7.1s (7/7) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                          0.0s
 => => transferring dockerfile: 175B                                                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                               2.2s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 2B                                                                               0.0s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 271B                                                                             0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e  4.6s
 => => resolve docker.io/library/nginx:latest@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e  0.0s
 => => sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e 10.23kB / 10.23kB                 0.0s
 => => sha256:fd204fe2f75024354b1f979d38cc43def9e049cc2df1cda45074d1b84c4f9b3e 9.09kB / 9.09kB                   0.0s
 => => sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419 29.78MB / 29.78MB                 1.0s
 => => sha256:b47f187216b68c3f90b7fb9a947a76d4bd2981a8f469cb6092e7da5d9efea014 33.14MB / 33.14MB                 1.3s
 => => sha256:1ad233904a1166fd3832d9657b163937f66c986795a517a87b516bc3f2481695 6268 / 6268                      1.0s
 => => sha256:b7f90eb243ff5e2c4a7dc61768160e808ac459e2c05aeb1b3bdf90920aa9d7e9 2.29kB / 2.29kB                   0.0s
 => => sha256:35ff83c394d65bf2c49f844ba18fe2e8121fe07e1f3f92325c05246319b98f7e 405B / 405B                       1.6s
 => => sha256:eedda9fd87862df4b2eccd3da9f2e38122124dec2e4217b530bd48d809b3840b 954B / 954B                       1.6s
 => => extracting sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419                       1.6s
 => => sha256:17d0911eaf62f5c4c34e50943d30dbb0d86b8ad614c80ef5ee02409ff54a3373e 1.21kB / 1.21kB                   1.9s
 => => sha256:df0b66c867e4916340d63f065ef5ea556cba20a97761039f9e9d5af714U2f8fbf 1.40kB / 1.40kB                   2.1s
 => => extracting sha256:b47f187216b68c3f90b7fb9a947a76d4bd2981a8f469cb6092e7da5d9efea014                       1.4s
 => => extracting sha256:1ad233904a1166fd3832d9657b163937f66c986795a517a87b516bc3f2481695                       0.0s
 => => extracting sha256:eedda9fd87862df4b2eccd3da9f2e38122124dec2e4217b530bd48d809b3840b                       0.0s
 => => extracting sha256:35ff83c394d65bf2c49f844ba18fe2e8121fe07e1f3f92325c05246319b98f7e                       0.0s
 => => extracting sha256:17d0911eaf62f5c4c34e50943d30dbb0d86b8ad614c80ef5ee02409ff54a3373e                       0.0s
 => => extracting sha256:df0b66c867e4916340d63f065ef5ea556cba20a97761039f9e9d5af714U2f8fbf                       0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                    0.1s
 => exporting to image                                                                                        0.0s
 => => exporting layers                                                                                       0.0s
 => => writing image sha256:2428b8a60c9cf46cbe89763514b534d388145b30683c4a6ee8bb7aee0a0aa06f                  0.0s
 => => naming to docker.io/ghostatdocker/my_nginx_image:1.0                                                   0.0s
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker images
REPOSITORY                  TAG       IMAGE ID       CREATED          SIZE
ghostatdocker/my_nginx_image   1.0       2428b8a60c9c   5 seconds ago   161MB
[ec2-user@ip-172-31-10-45 my_nginx_image]$
[ec2-user@ip-172-31-10-45 my_nginx_image]$
```

Logging in to Docker and pushing the custom image into it:

```
ec2-user@ip-172-31-10-45:~/i
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to
create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using
 SSO. Learn more at https://docs.docker.com/go/access-tokens/

Username: ghostatdocker
Password:
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[ec2-user@ip-172-31-10-45 my_nginx_image]$ docker push ghostatdocker/my_nginx_image:1.0
The push refers to repository [docker.io/ghostatdocker/my_nginx_image]
3bc00d44e172: Pushed
87026f913c5d: Mounted from library/nginx
d120c727e5e3: Mounted from library/nginx
a821e7df95cb: Mounted from library/nginx
b9ae255e9e51: Mounted from library/nginx
a507d5bc786a: Mounted from library/nginx
7351f826b314: Mounted from library/nginx
a257f20c716c: Mounted from library/nginx
1.0: digest: sha256:da62c79cbcd3ed4f8e7022d853e58a7e52c398bc9ace0116b072ab0952c1f265 size: 1985
[ec2-user@ip-172-31-10-45 my_nginx_image]$
```